

WINADRYL EXPECTORANT SYRUP

(Antitussive - Antihistamine - Expectorant)

**Active Ingredients:**  
Each 5ml contains -  
Diphenhydramine HCl ..... 14mg  
Ammonium Chloride ..... 135mg  
Preserved with Methyl Hydroxybenzoate 0.12%w/v and Propyl Hydroxybenzoate 0.012%w/v.

**Presentations:**  
Bottles of 90ml, 100ml and 120ml.

**Product Descriptions:**  
Dark brown coloured syrupy liquid with plum flavour.

**Pharmacology:**  
Diphenhydramine, a monoethanolamine derivative, is an antihistamine with antimuscarinic and sedative properties. It is commonly used for symptomatic relief of cough and other symptoms associated with common cold. Ammonium Chloride has an irritant effect on mucous membranes and is considered to have expectorant properties. Diphenhydramine Hydrochloride is well absorbed from the gastrointestinal tract, although it appears to have a high first-pass metabolism. Peak plasma concentrations are achieved about 1 to 4 hours after oral administration. Diphenhydramine is widely distributed throughout the body including the CNS. Diphenhydramine is highly bound to plasma proteins. Diphenhydramine is excreted mainly in the urine as metabolites and very little is excreted as unchanged drug. The elimination half-life has been reported to range from 2 to 9 hours. Ammonium Chloride is absorbed from the gastrointestinal tract. The ammonium ion is converted into urea in the liver; the anion thus liberated into the blood and extracellular fluid causes a metabolic acidosis and decreases the pH of the urine; this is followed by transient diuresis.

**Indications:**  
For the relief of cough and congestion in the chest and nasal passages. It helps to loosen viscous mucous and expectorate stubborn phlegm.

**Dosage and Administration:** For oral administration only.  
Adults and children above 12 years:  
Take 1 to 2 teaspoons (5 to 10ml) every 4 to 6 hours.  
Children 6 to 12 years:  
Give ½ to 1 teaspoon (2.5 to 5ml) every 4 to 6 hours.  
Children 2 to 5 years:  
Give ¼ teaspoon (2.5ml) every 4 to 6 hours.  
Do not exceed 6 doses in 24 hours unless advised by a doctor.

**Symptoms and Treatments of Overdose:**  
In any cases of overdose, please call your doctor or seek emergency medical attention immediately.  
*Diphenhydramine HCl*  
Symptoms: Overdosage with sedating antihistamines is associated with antimuscarinic, extrapyramidal, and CNS effects. When CNS stimulation predominates over CNS depression, which is more likely in children or the elderly, it causes ataxia, excitement, tremors, psychoses, hallucinations, and convulsions; hyperpyrexia may also occur. In adults, CNS depression is more common with drowsiness, coma, and convulsions, progressing to respiratory failure and cardiovascular collapse.  
Treatments: Treatment should be supportive and directed towards specific symptoms. Measures to promote rapid gastric emptying (with Ipecac-induced emesis or gastric lavage) and in cases of acute poisoning, activated charcoal may be used. Convulsions and marked CNS stimulation should be treated with parenteral Diazepam.

**Ammonium Chloride**  
Symptoms: Ammonium salts are irritant to the gastric mucosa and may produce nausea and vomiting particularly in large doses. Large doses of Ammonium Chloride may cause a profound acidosis and hypokalaemia which should be treated symptomatically.  
Treatments: Treatment consists of the correction of acidosis and electrolyte loss by the intravenous administration of sodium bicarbonate or sodium lactate, together with potassium supplements by mouth. Otherwise treatment should be symptomatic and supportive.

**Contraindication:**  
Not suitable for patients who have known hypersensitivity to Diphenhydramine and Ammonium.

**Precautions:**  
If symptoms persist, please consult a doctor.  
***This medicine may cause drowsiness. If affected, do not drive or operate machinery. Avoid alcoholic drinks.***  
It should be used with caution in patients with hepatic or renal impairment. Not to be used in children less than 2 years of age.  
To be used with caution and doctor's / pharmacist's advice in children 2 to 6 years of age.

**Use in Pregnancy and Lactation:**  
The safety of this product during pregnancy has not been established, although there is no good evidence of an association between first trimester exposure to either ammonium chloride or diphenhydramine and foetal abnormalities. The safety of the product during lactation has not been established, although use during this period would not be considered to constitute a hazard.

**Interactions with Other Medicaments:**  
Sedating antihistamines, such as Diphenhydramine may enhance the sedative effects of CNS depressants including alcohol, barbiturates, hypnotics, opioid analgesics, anxiolytic sedatives, and antipsychotics. Sedating antihistamines have an additive antimuscarinic action with other antimuscarinic drugs, such as Atropine and some antidepressants (both tricyclics and MAOIs). Diphenhydramine inhibits the cytochrome P450 isoenzyme CYP2D6 that is partly responsible for the metabolism of some beta blockers including Metoprolol and the antidepressant Venlafaxine.

**Undesirable Effects:**  
The most common adverse effect is CNS depression, with effects varying from slight drowsiness to deep sleep, and including lassitude, dizziness, and incoordination. These sedative effects, when they occur, may diminish after a few days of treatment. Other adverse effects that are more common include headache, psychomotor impairment, and antimuscarinic effects, such as dry mouth, thickened respiratory-tract secretions, blurred vision, urinary difficulty or retention, constipation, and increased gastric reflux. Occasional gastrointestinal adverse effects of antihistamines include nausea, vomiting, diarrhoea, or epigastric pain.

**Storage:**  
Store below 30°C. Protect from light. Keep cap tightly closed.  
Keep out of reach of children. Jauhkan daripada capaian kanak-kanak.

**Shelf Life:**  
3 years from the date of manufacture.

Manufacturer and Product Registration Holder  
**WINWA MEDICAL SDN. BHD.**  
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**MAL08010745AZ BRU24063455P**

Size: 75mm x 135mm

Dark Blue (Pantone 2767U)